FGAI Assignment 3

English to French Machine Translation using Transformer mT5 Llama and Gemma

This submission provides the repository for a four-model machine-translation study using a shared English-to-French test set. Llama 3.2 1B achieved the best measured translation quality in the completed experiment.

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Assignment title	Foundations of Generative AI Assignment 3
System title	Four Architecture English to French Machine Translation System
Dataset	OPUS Books English French parallel corpus https://huggingface.co/datasets/Helsinki-NLP/opus_books
GitHub repository	Sachithvp fgai assignment3 machine translation https://github.com/Sachithvp/fgai-assignment3-machine-translation

The repository includes the runnable notebook and Python implementation, fixed data splits, measured BLEU and chrF results, timing and parameter comparisons, temperature analysis, plots, and generated translation samples.